

Cambridge (CIE) IGCSE Maths
Paper 1: Non-Calculator (Set D)
Core

Saturday 26 April 2025

Morning (Time: 1 hour 30 minutes)

Total marks

/ 80

Instructions

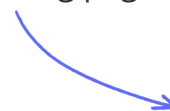
- Try to complete this mock exam paper in one sitting, under exam conditions. Use all the time available and check your answers to each question at the end before submitting.
- Remember this is PRACTICE. Mistakes are fine and will help you improve in time for the real exam - just do your best.
- You will need geometrical instruments.
- Calculators must not be used in this paper.

Materials

- [List of formulas](#)

Scan here to mark your mock exam

or visit the mock exams landing page for this course



1 (a) The attendance at a football match is 11 678.

Write 11 678 in words.

[1]

(1 mark)

(b) Write 11 678 correct to the nearest 100.

(1 mark)

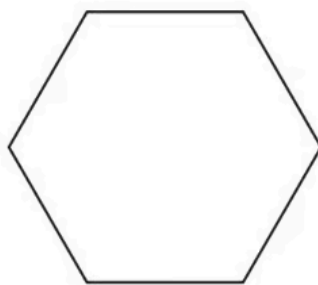
2 When Marco arrives at a port the temperature is 5°C .
At midnight the temperature has fallen by 7°C .

Find the temperature at midnight.

..... $^{\circ}\text{C}$

(1 mark)

3 (a) The diagram shows a regular polygon.



Write down the mathematical name for this shape.

(1 mark)

(b) Write down the order of rotational symmetry of this shape.

(1 mark)

4 Find

i) the smallest multiple of 7 that is greater than 100,

[1]

ii) the largest cube number that is less than 100,

[1]

(2 marks)

5 Complete the table.

Fraction		Decimal		Percentage
$\frac{3}{4}$	=	0.75	=	
	=	0.2	=	20%
$\frac{2}{25}$	=		=	8%

[3]

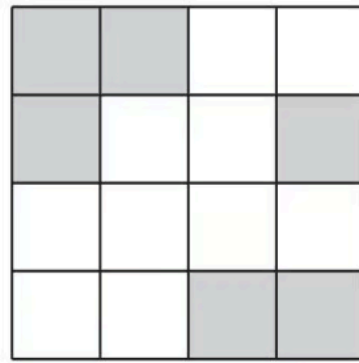
(3 marks)

- 6** When full, a cruise ship carries 880 guests and 360 crew.

Write the ratio guests : crew in its simplest form.

(1 mark)

7



Shade two squares so that the diagram has rotational symmetry of order 2.

(1 mark)

8 The scale drawing shows a rock, R .

The scale is 1 centimetre represents 30 metres.

A lighthouse, L , is 210m from R , on a bearing of 125° .

On the diagram, mark the position of L .



Scale : 1 cm to 30 m

(2 marks)

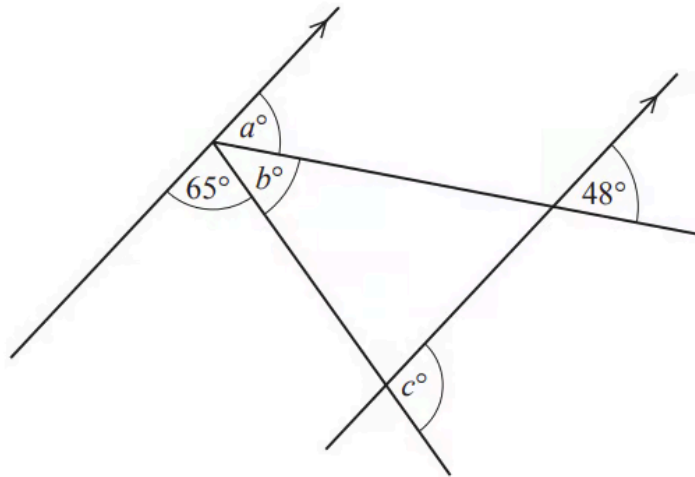
9 There are 12 brass instruments in an orchestra.

$\frac{1}{3}$ are trumpets, 3 are trombones and the remainder are horns.

Find the number of horns.

(2 marks)

10 (a)



NOT TO
SCALE

The diagram shows two parallel lines and two straight lines.

Find the value of a .

Give a reason for your answer.

$a = \dots\dots\dots$ because $\dots\dots\dots$

(2 marks)

(b) Find the value of b .

Give a reason for your answer.

$b = \dots\dots\dots$ because $\dots\dots\dots$

(2 marks)

(c) Find the value of c .

$c = \dots\dots\dots$ [2]

(2 marks)

11 (a) Calculate $3 + 2 \times (8 - 6)$

(1 mark)

(b) Find the value of

$$3^0,$$

(1 mark)

(c) 5^{-2} .

(1 mark)

12 (a) Factorise $12x + 15$

(1 mark)

(b) Solve.

$$7x - 3 = 2x + 11$$

$x = \dots\dots\dots$

(2 marks)

13 Diego flies from Madrid to Buenos Aires.

His flight leaves at 20 55 and arrives at 03 50 local time.

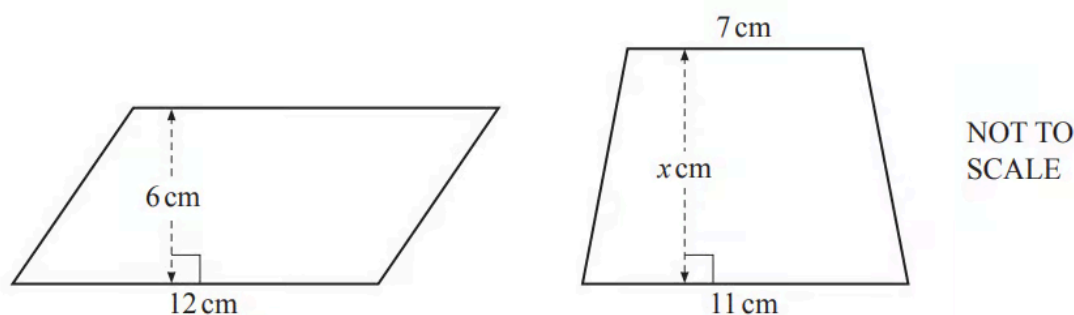
The local time in Buenos Aires is 5 hours behind the local time in Madrid.

Work out, in hours and minutes, the time the flight takes.

$\dots\dots\dots$ h $\dots\dots\dots$ min

(2 marks)

14



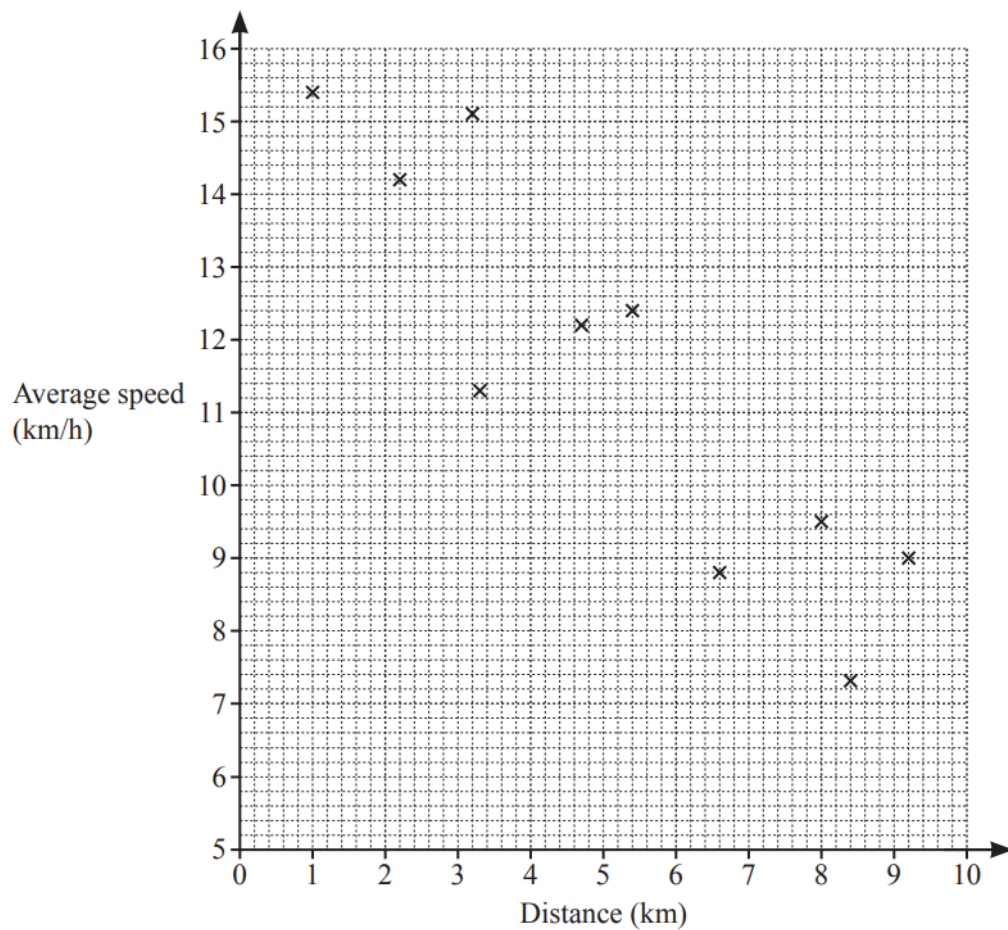
The area of the parallelogram is the same as the area of the trapezium.

Work out the value of x .

$x = \dots\dots\dots$

(3 marks)

- 15 (a)** Aisha records the distance she runs and her average speed. The results are shown in the scatter diagram.



The table shows the results of four more runs.

Distance (km)	4.2	5.7	7.1	8.8
Average speed (km/h)	13.4	11.8	9.8	8.3

On the scatter diagram, plot these points.

(2 marks)

(b) What type of correlation is shown in the scatter diagram?

(1 mark)

(c) On the scatter diagram, draw a line of best fit.

(1 mark)

(d) Use your line of best fit to estimate her average speed when she runs a distance of 6km.

.....km/h

(1 mark)

16 The table shows the maximum power, kW, and the time taken, in seconds, to accelerate from 0 to 100 km/h for each of 10 cars.

Maximum power (kW)	77	52	103	55	44	51	85	135	90	110
Time (seconds)	12.5	14.9	9.0	12.1	14.4	12.9	10.0	7.1	11.0	9.4

i) Find the range of the times.

..... s [1]

ii) Find the median maximum power.

..... kW [2]

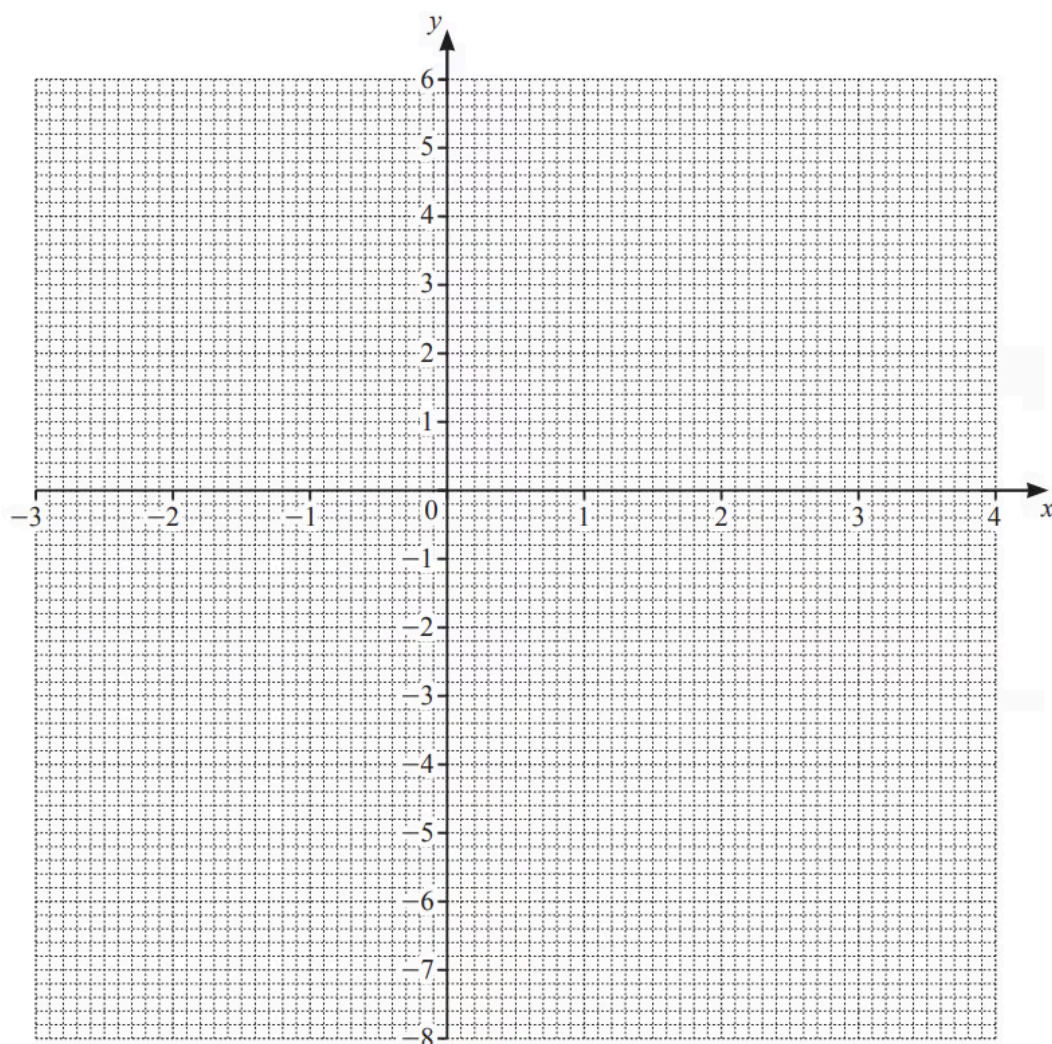
(3 marks)

17 (a) Complete the table of values for $y = -x^2 + x + 5$

x	-3	-2	-1	0	1	2	3	4
y		-1	3			3		

(3 marks)

(b) On the grid, draw the graph of $y = -x^2 + x + 5$ for $-3 \leq x \leq 4$.



(4 marks)

(c) Write down the coordinates of the highest point of the graph.

(1 mark)

(d) Write down the equation of the line of symmetry of the graph.

(1 mark)

(e) i) On the grid, draw the line $y = x$ for $-3 \leq x \leq 4$

[1]

ii) Write down the values of x where the line $y = x$ crosses the curve $y = -x^2 + x + 5$

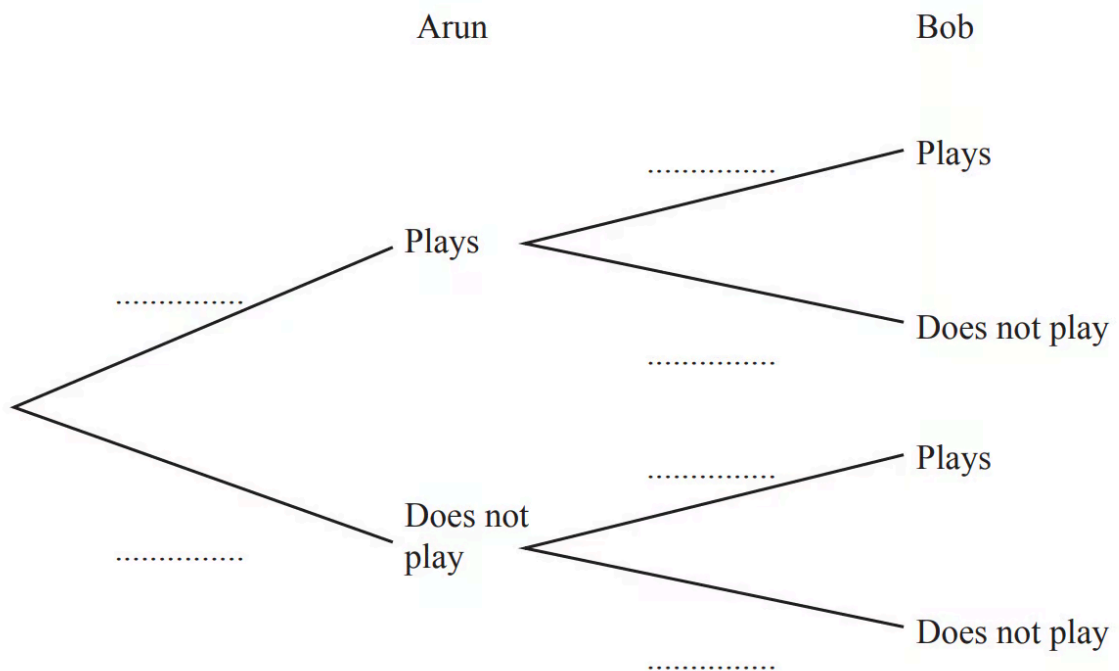
$x = \dots\dots\dots$ and $x = \dots\dots\dots$ [2]

(3 marks)

18 On any Saturday, the probability that Arun plays football is $\frac{3}{4}$.

On any Saturday, the probability that Bob plays football is $\frac{2}{5}$.

i) Complete the tree diagram.



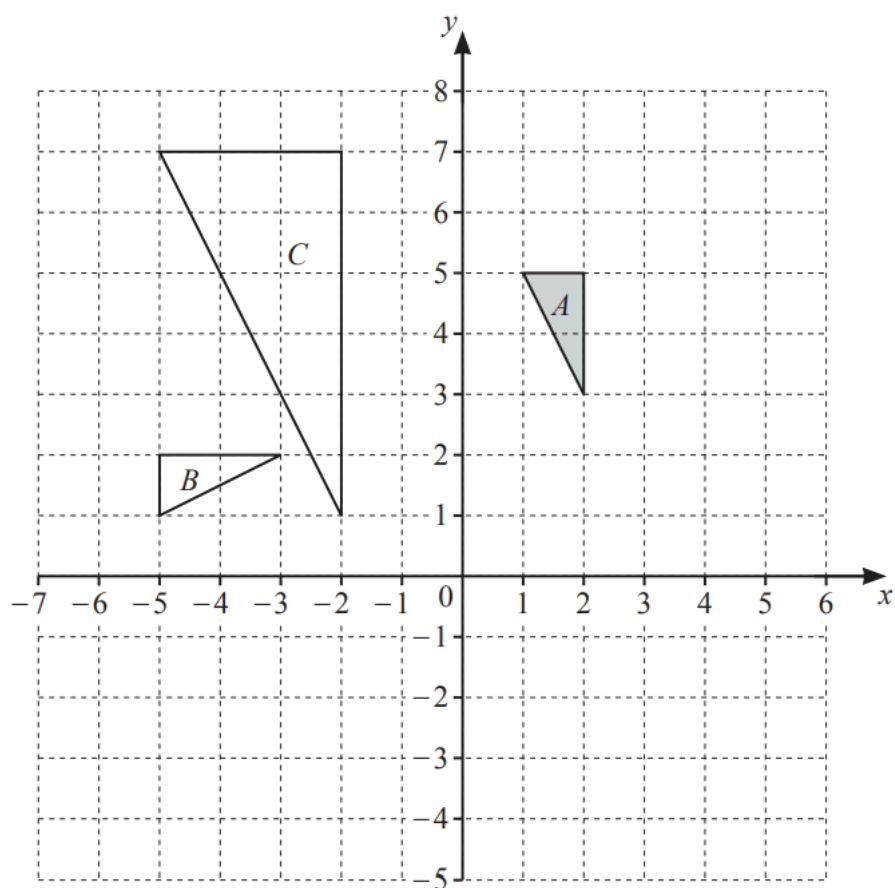
[2]

ii) Calculate the probability that, one Saturday, Arun and Bob both play football.

[2]

(4 marks)

19 (a)



On the grid, draw the image of

ia) triangle A after a translation by the vector $\begin{pmatrix} 3 \\ -7 \end{pmatrix}$,

[2]

ib) triangle A after a reflection in the line $x = 3$.

[2]

(4 marks)

(b) ii) Describe fully the **single** transformation that maps triangle A onto triangle B .

[3]

iii) Describe fully the **single** transformation that maps triangle A onto triangle C .

[3]

(6 marks)

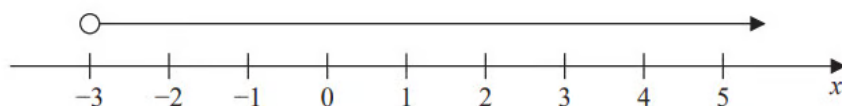
20 Make r the subject of this formula.

$$p = 3r - 5$$

$$r = \dots\dots\dots$$

(2 marks)

21 (a)



Write down the inequality shown on the number line.

(1 mark)

(b) List all of the integers which satisfy **all three** of the following inequalities.

$$x \leq 8$$

$$x > 2$$

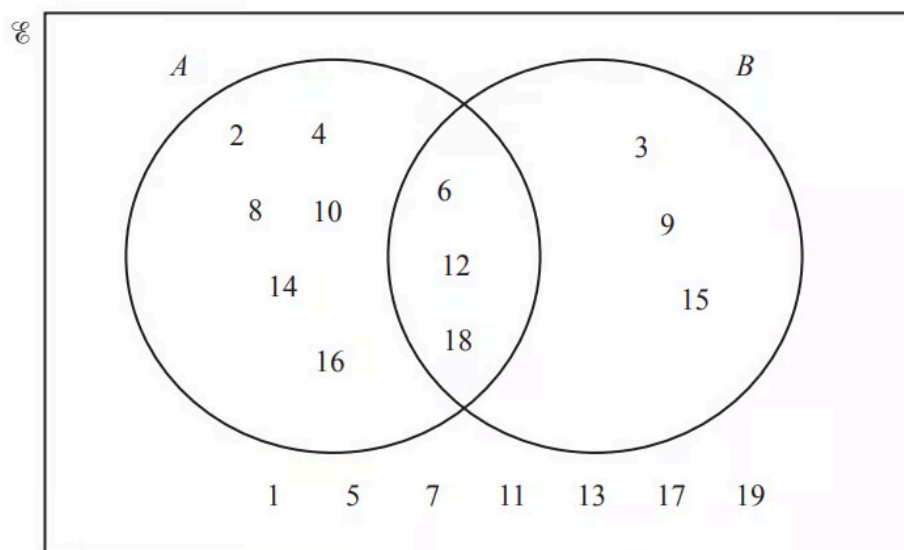
$$x > 3$$

(1 mark)

22 (a) $\mathcal{E} = \{x : x \text{ is a positive integer less than } 20\}$

$A = \{x : x \text{ is an even number}\}$

$B = \{x : x \text{ is a multiple of } 3\}$



Write down $n(A)$.

[1]

(1 mark)

(b) List the elements of set B .

$B = \{ \dots\dots\dots \}$

(2 marks)

(c) Complete the statement.

$A \cap B = \{x : x \text{ is } \dots\dots\dots\}$

(1 mark)

- 23** The height, h metres, of a tower is 76.3 m, correct to 1 decimal place.
Complete this statement about the value of h .

..... $\leq h <$

(2 marks)

- 24** Solve the simultaneous equations.

You must show all your working.

$$5x - 2y = 26$$

$$7x + 6y = 10$$

$x =$

$y =$

(3 marks)